Electrical Power and Energy Systems 146 (2023) 108783

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Detection of the stability boundary associated to subsynchronous oscillations in DFIG based wind farms

Nicolás E. Costa [a,*], Gustavo Revel [b], Diego M. Alonso [b], Roberto D. Fernández [a]

[a] Facultad de Ingeniería, Universidad Nacional de la Patagonia San Juan Bosco, Ruta 1 S/N Ciudad Universitaria, (9000) Comodoro Rivadavia, Argentina
[b] Instituto de Investigaciones en Ing. Eléctrica (IIIE) Alfredo Desages (UNS-CONICET), Dpto. de Ing. Eléctrica y de Computadoras, Universidad Nacional del Sur San Andrés 800, (8000) Bahía Blanca, Argentina

ARTICLE INFO

Keywords:
Subsynchronous oscillations
Doubly-fed induction generator
Stability boundary
Selective Modal Analysis

ABSTRACT

In this paper, the detection and continuation of the Hopf bifurcation defining the stability boundary associated to subsynchronous oscillations (SSO) in DFIG-based wind farms is considered. In order to overcome the main limitation of standard numerical continuation packages for studying SSO in large scale power systems, a simplified test function, based only on the subsynchronous and supersynchronous modes, is proposed. These modes are computed with an adapted version of the Selective Modal Analysis (SMA) iterative method, which avoids the computation of all the eigenvalues of the system. The iterative procedure used for the detection of the Hopf bifurcation and the standard prediction–correction routines used for its continuation are tailored to accommodate both the proposed test function and the SMA based computation of the required eigenvalues. The methodology is illustrated on a relatively large scale system with 11 WF represented by 22 aggregated models and involving 690 state variables.

1. Introduction

Subsynchronous oscillations (SSO) involving doubly feed induction generator (DFIG) based wind farms (WF) and series compensated transmission lines are characterized by the emergence of poorly damped oscillations with frequency below the synchronous frequency. These oscillations can cause damage to the system equipment (wind turbines, series compensation, etc.) and tripping of wind turbines, among others. The first incidents occurred more than a decade ago [1–3] and they were followed by several events, mainly in WF located in China and USA [4,5]. These incidents captured the attention of researchers, transmission system operators (TSO) and manufacturers, triggering the development of analysis methods and procedures for SSO risk assessment for transmission expansion projects and generation interconnection [6,7]. The analysis methods include eigenvalue or modal analyses considering state-space models [8–14], frequency responses obtained from simulations on electromagnetic transient (EMT) models or from actual measurements [15–23], or a combination of them [24,25]. Simultaneously, countermeasures and mitigation techniques were developed, such as those based on introducing damping through flexible AC transmission systems (FACTS), modifying wind turbine controls, or including special purpose converters or suppression filters, among others (see the compilation [26]).

The phenomenon is inherently electrical and does not involve the torsional dynamics of the turbine-generator assembly. It is related to the interaction between the DFIG and the series compensation capacitor, and it is physically explained by the induction generator effect (IGE) caused by the negative resistance introduced by the wind turbine generator at the subsynchronous frequency [27]. Small-signal eigenvalue analysis reveals that the oscillation is associated to a pair of complex conjugate eigenvalues, the so-called subsynchronous mode, which appears due to this interaction. The damping and frequency of the subsynchronous mode, and therefore the system stability, is affected by several parameters such as the wind speed, the compensation level, the number of wind turbines in operation, the line impedances, the wind turbine control schemes and the controller tuning [4]. When some of these parameters vary such that the damping of the subsynchronous mode is reduced and it crosses the imaginary axis, the system loses stability. Therefore, the set of parameter values leading to this critical operating condition defines the stability boundary where the subsynchronous mode has zero real part or, equivalently, the net resistance of the system at the subsynchronous frequency is zero. Detecting this boundary to predict the risk of triggering an undamped SSO event, is particularly useful in complex scenarios, such as those with multiple and geographically distributed WF [4,12]. In general, this task requires

* Corresponding author.
E-mail addresses: ncosta@unpata.edu.ar (N.E. Costa), grevel@uns.edu.ar (G. Revel), dalonso@uns.edu.ar (D.M. Alonso), dfernandez@unpata.edu.ar (R.D. Fernández).

https://doi.org/10.1016/j.ijepes.2022.108783
Received 2 April 2022; Received in revised form 29 September 2022; Accepted 7 November 2022
Available online 23 November 2022
0142-0615/© 2022 Elsevier Ltd. All rights reserved.

the evaluation of a significant number of operating conditions, which is difficult and challenging for most of the analysis methods. In practice, the impedance model analysis is the most widely used method because the model can be derived from experimental data on devices whose internal structure and parameters are not completely known. Since the impedance model represents the behavior at a specific operating condition, the method has been adapted to consider variations of the operating point [28,29] or to obtain aggregated impedance models of a WF based on the impedance model of individual wind turbines [30]. Current research is focused on extending the approach to identify the critical operating conditions that define the stability boundary. In this regard, in [31] the impedance network model of the power system is constructed by connecting the impedance model of the different components of the system, and obtaining the corresponding aggregated impedance. The frequency response of the aggregated impedance is used to detect, via a step-wise search method, the critical conditions delimiting the security region. In [32], converter interfaced generators are considered, and the boundary is obtained by considering as variable parameters the output power of the converter (represented by a black-box model) and the short-circuit ratio at the point of common coupling. Then, the so called variable-operating-point impedance model of the converter is evaluated, and the stability boundary corresponding to the zero damping condition is obtained by the generalized Nyquist criterion. Due to the great difficulty to directly solve the zero damping condition, a step-wise search method is implemented by fixing the output power at different values in the feasible range. In [33], the variable-operating-point impedance model approach is used to identify the stability boundary of a grid-tied voltage source converter. The method considers as variable parameters the active and reactive powers of the converter and defines a mesh of discrete values for the parameters where the stability condition is evaluated, provided that the point satisfies some operating constraints. As in the previous case, the stability boundary is obtained for the full range of the considered operating conditions. The impedance concept was also used to obtain critical conditions in order to avoid resonances in WF considering the positive-net-damping stability criterion [34] and the harmonic voltage amplification [35].

Another commonly used approach is the eigenvalue or modal analysis. This approach is based on the state-space representation of the model which requires the knowledge of the internal structure and the parameters of the wind turbine, which are rarely provided by manufacturers. On the other hand, the state-space representation inherently contains information about the dynamical behavior of the system and, thus, allows to go further in the theoretical understanding of the phenomenon. The state-space representation is considered in [36] to derive a method based on Gerschgorin disk theorem to find the parameter stability region which serves as a guidance for selecting the controller to prevent and control SSO for a given operating point. In addition, in [37], the security region method is presented as an alternative to the general point-wise method for detecting the critical conditions. Focusing the attention on the subsynchronous mode, the critical or zero damping condition occurs when this mode crosses the imaginary axis and, according to the bifurcation theory, the mechanism is known as a Hopf bifurcation [38]. Therefore, the locus of the Hopf bifurcation in the parameter space defines the stability boundary associated to the SSO phenomenon [12]. In addition, the detection and continuation of the stability boundary, defined by the Hopf bifurcation, can be done using standard software packages such as MatCont [39]. Nevertheless, these routines are not designed to handle systems with hundreds (or even thousands) of state variables such as those arising in power system modeling. The main restriction resides in the evaluation of the so called test function, a scalar function that vanishes at the critical condition, which is used to verify that the equilibrium, or operation point, undergoes a bifurcation. To overcome this problem, in this paper the critical condition is verified using a test function based only on the subsynchronous and supersynchronous modes. These modes are computed by means of the Selective Modal Analysis (SMA) method [40,41] which avoids the computation of all the eigenvalues and eigenvectors of the system. This is an iterative method and its convergence is highly dependent on the selection of a subset of the state variables, named relevant variables. This issue was addressed in [42], where a systematic two-phase procedure for reproducing the subsynchronous and supersynchronous modes via the SMA method using the minimum number of relevant variables was proposed. Therefore, the main contribution of this paper, consists in developing a systematic methodology for computing the stability boundary associated to the SSO phenomenon. The proposed methodology exploits the strength of the continuation methods to detect, locate and trace the critical condition (using the information of the state-space model of the system). The advantage of this approach is that, once a critical condition is located, the algorithm tracks it in the parameter space by varying not only the parameters but also the operating point. In addition, the two-phase SMA approach is embedded in the numerical continuation algorithm to provide only the four eigenvalues of interest (super and subsynchronous modes) needed for evaluating the test function. This approach not only avoids the computation of all the eigenvalues of the system but also allows the implementation of the continuation strategy to track the critical boundary on large scale systems. The methodology is illustrated on a relatively large scale system with 11 WF represented by 22 aggregated models and involving 690 state variables.

The rest of the paper is organized as follows. In Section 2, some preliminary concepts on the phenomenon are presented, and the continuation routines are described in Section 3. The test function and the procedure for its evaluation is included in Section 4. In Section 5, the proposed methodology is illustrated using a prototypical power system. Finally, in Section 6 some conclusions are presented.

2. Preliminary concepts

Subsynchronous oscillations involving DFIG-based WF and series compensation is an electromagnetic phenomenon and thus the model of the power system must be capable of reproducing the dynamic behavior in this time scale. To describe the general approach followed to model the power system and to introduce the phenomenon, the power system of Fig. 1 is considered. The network resembles the southern portion of the Argentinian Interconnection System. It is composed by 29 buses with a 500 kV corridor (in red) connected, at bus 1008, to an equivalent system represented by the corresponding short circuit impedance and an ideal power source. The parameters of the network, including line lengths as well as transformers and line impedances, are obtained from the TSO and can be consulted in [42], Table 10. The first section of the 500 kV transmission system (between buses 1008 and 11) has a series compensation capacitor bank and the compensation level of that line is represented by the reactance ratio $\mu = X_c/X_{1008-11}$ $(0 < \mu < 1)$. The eleven DFIG-based WF depicted in Fig. 1 are composed by a different number of modules ranging from 20 to 75 MW, as described in Table 1 (all the parameters are obtained form the TSO). To simplify the later analysis, the WF are arranged into four clusters, namely PM1, PM2, CR and SCN, as shown in Fig. 1 and Table 1.

2.1. Models of the power system and WF

The lines are represented by the corresponding π models considering the line currents and bus voltages as state variables. The WF are represented by aggregated models, including an equivalent wind turbine with the corresponding back-to-back converters in the rotor circuit and their control schemes, and the equivalent internal network for each WF module. All the electrical variables are defined in the d-q frame. The model is obtained following the standard approach described in [6] and the details on the equivalent wind turbine model, controller structure and parameters are described in [42].

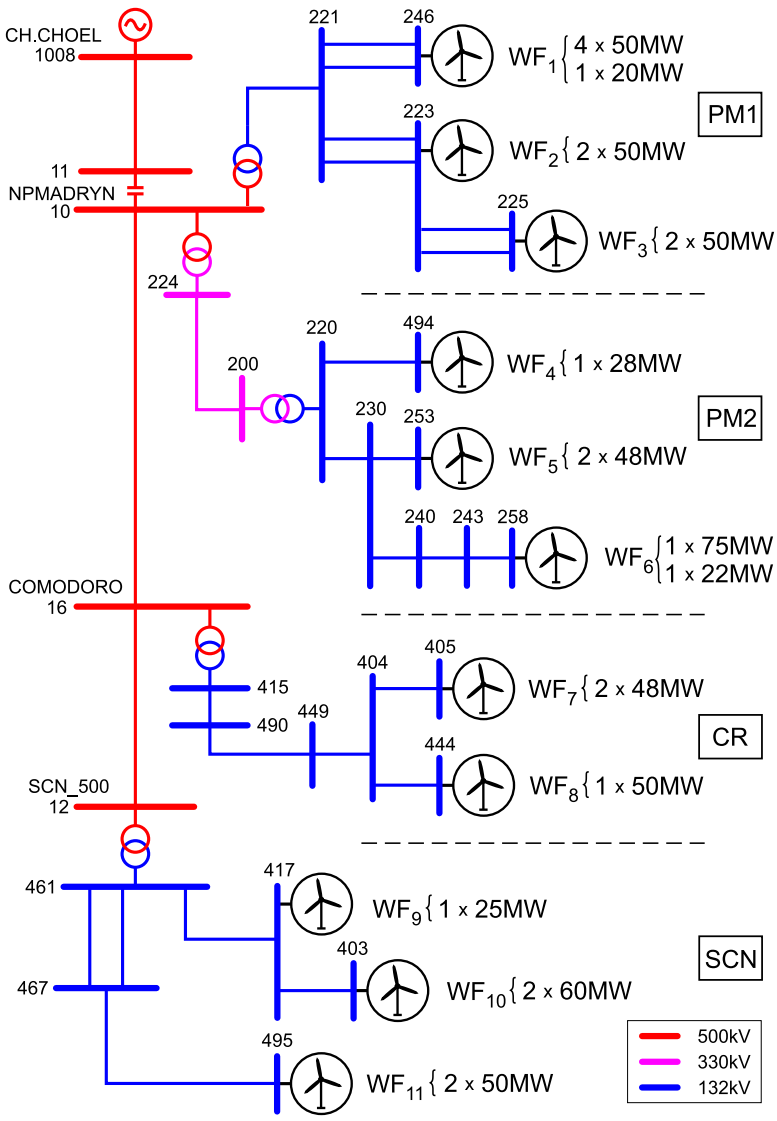


Fig. 1. Single line diagram of the prototypical power system used in this paper to illustrate the proposed methodology.

The power system model results in a set of nonlinear ordinary differential equations of the form

$$\dot{x} = f(x, p), \tag{1}$$

where $f : \mathbb{R}^N \times \mathbb{R}^P \to \mathbb{R}^N$ is a smooth vector field in (x, p), $x \in \mathbb{R}^N$ is the state vector and $p \in \mathbb{R}^P$ is the parameter vector. In this case, it results $N = 690$ states, where 206 of them correspond to bus voltages and line currents (including the equivalent internal network of each WF module) and the remaining 484 states correspond to the 22 WF modules (22 states per equivalent wind turbine).

2.2. Small signal analysis

The SSO phenomenon can be studied by analyzing the behavior of the subsynchronous mode of the linearization matrix of the power system (1) at a given operation point. This oscillation mode is associated to the series compensation capacitor which introduces a resonance condition below the synchronous frequency. Due to the physical coupling at the generator windings and the model representation using a synchronous d-q frame, the introduced resonance frequency gives place not only to the subsynchronous mode but also to the supersynchronous mode at complimentary frequencies around the synchronous frequency. Generally, the damping of the subsynchronous mode is affected by the operating condition. For example, when the compensation level is increased, the resonance frequency increases and the slip of the machine at the subsynchronous frequency, which is negative at this frequency, is reduced. Since the equivalent DFIG rotor resistance is inversely proportional to the slip, the negative value increases and this reduces the equivalent resistance of the DFIG seen from the stator terminals, paving the way to the emergence of poorly damped SSO [4]. A reduction in the wind speed, also reduces the slip at the subsynchronous frequency. This reduces the equivalent resistance at the stator of the DFIG, and thus the stability margin is also diminished. On the other hand, the behavior of the supersynchronous mode is usually opposite to that of the subsynchronous one. The damping is also affected by the tuning of the WF controller, the number of wind turbines in operation and the impedance of the lines, among others. Ultimately, when the operating condition changes, due to a parameter variation or a network reconfiguration, and the subsynchronous mode crosses to the right-hand side of the complex plane, the system loses stability due to a Hopf bifurcation. Thus, the locus of the Hopf bifurcation in the parameter space defines the stability boundary associated to this phenomenon.

To illustrate the problem, the subsynchronous and supersynchronous modes corresponding to the system in Fig. 1 are computed at different operating conditions. The eigenvalues resulting from varying the compensation level μ from 10% to 90% (with steps of 2%) are

Table 1
Power of WF modules and equivalent impedances (values in p.u. for a base power of 100 MVA and nominal voltages).

Cluster	WF	Module	33/132 kV	Internal network			0.69/33 kV
(Power)		power	X_T	R	X	B	X_T
PM1 (4.20)	1	0.20	0.2000	0.03853	0.03812	0.0048	0.1607
		0.50	0.2000	0.01645	0.01667	0.0088	0.1607
		0.50	0.2000	0.01546	0.01581	0.0097	0.1607
		0.50	0.2000	0.01959	0.02004	0.0102	0.1607
		0.50	0.2000	0.01496	0.01499	0.0094	0.1607
	2	0.50	0.1250	0.01047	0.01896	0.0489	0.1800
		0.50	0.1250	0.01357	0.01345	0.0048	0.1800
	3	0.50	0.1250	0.01008	0.01652	0.0221	0.1800
		0.50	0.1250	0.00861	0.01949	0.0024	0.1800
PM2 (2.21)	4	0.28	0.2000	0.00861	0.01949	0.0024	0.1524
	5	0.48	0.2070	0.01160	0.01000	0.0075	0.1143
		0.48	0.2070	0.01160	0.01000	0.0075	0.1143
	6	0.22	0.1833	0.00422	0.00456	0.0129	0.8394
		0.75	0.1833	0.00422	0.00456	0.0129	0.8394
CR (1.46)	7	0.48	0.1830	0.01160	0.01000	0.0075	0.1600
		0.48	0.1830	0.01160	0.01000	0.0075	0.1600
	8	0.50	0.1730	0.01645	0.01667	0.0088	0.1818
SCN (2.45)	9	0.25	0.3330	0.01160	0.01000	0.0075	0.2857
	10	0.60	0.1700	0.01160	0.01000	0.0075	0.1428
		0.60	0.1700	0.01160	0.01000	0.0075	0.1428
	11	0.50	0.1830	0.01645	0.01667	0.0088	0.1636
		0.50	0.1830	0.01645	0.01667	0.0088	0.1636

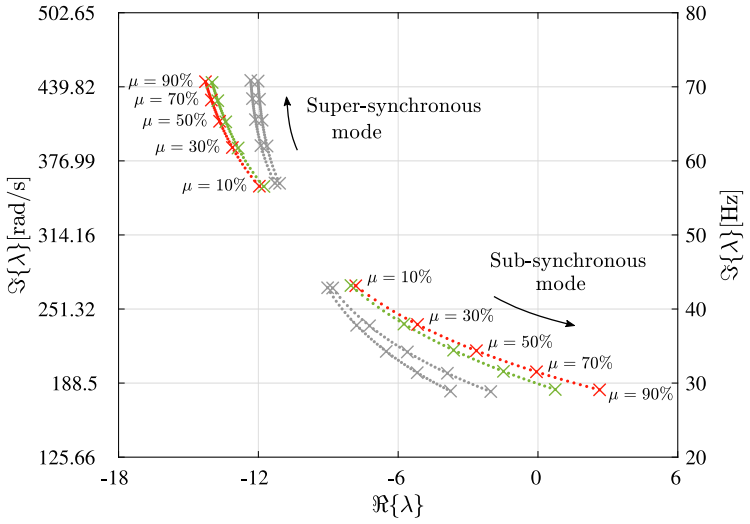


Fig. 2. Subsynchronous and supersynchronous modes (positive imaginary part only) for compensation levels μ from 10% to 90%, wind speeds fixed at $v_w = 5.5$ m/s (red) and 10.5 m/s (green) and proportional gain of the inner current loop of $K_r = 0.044$ (the ones for $K_r = 0.022$ are shown in gray).

shown in Fig. 2 (the complex conjugates are omitted for simplicity) for two different wind speeds: $v_w = 5.5$ m/s (red) and $v_w = 10.5$ m/s (green). The damping of the subsynchronous mode is reduced when the compensation level is increased, losing the stability at $\mu \simeq 70\%$ for a wind speed of 5.5 m/s (see the red markers) and at $\mu \simeq 83\%$ for a wind speed of 10.5 m/s (green markers). It is clear that the problem is aggravated in low wind speed regimes. Fig. 2 also shows (in gray) the eigenvalues resulting when the gain of the inner current loop of the rotor side converter controller in all the WF models is halved (from $K_r = 0.044$ to $K_r = 0.022$). In general, a higher gain (bandwidth) of the current control loop deteriorates the damping and jeopardizes the stability. Notice that, in all the cases, the damping and frequency of the supersynchronous mode increase when those of the subsynchronous mode decrease, i.e. they present opposite behaviors.

In this example, the computation of each equilibrium point at the different operating conditions (compensation level, wind speed and control gains) is carried out by increasing the parameter with a given step-size and solving numerically the equilibrium equation via the Newton–Raphson (NR) method. This kind of "brute force" approach is simple but presents limitations, mainly for computing the locus of the Hopf bifurcation in the parameter space. These tasks can be carried out more appropriately using the information provided by the vector field, for example, by means of the numerical continuation methods described in the following section.

3. Numerical continuation of equilibrium points and Hopf bifurcations

In order to describe the procedure for the numerical continuation of equilibrium points and Hopf bifurcations, the power system model (1) is expressed in terms of the so called continuation parameters, i.e. a subset α of the parameters p that are varied according to the continuation problem at hand (without loss of generality, the remaining parameters are left fixed). Thus, the system (1) results

$$\dot{x} = f(x, \alpha), \tag{2}$$

where $\alpha \subseteq p$ $(\alpha \in \mathbb{R}^K)$ and $f : \mathbb{R}^N \times \mathbb{R}^K \to \mathbb{R}^N$. For the following development, it is convenient to define $y = (x, \alpha) \in \mathbb{R}^{N+K}$ and $F(y) =$

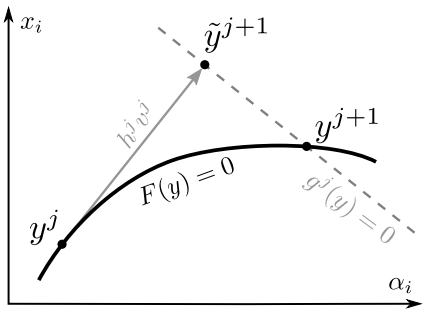


Fig. 3. Pseudo-arclength continuation scheme for equilibrium point continuation.

$f(x,\alpha)$, where $F : \mathbb{R}^{N+K} \to \mathbb{R}^N$. The continuation procedure is well known (see, for example, [43]) and the standard approach is briefly described below.

3.1. Continuation of the equilibrium point

The continuation of the equilibrium point is performed by varying a single parameter, i.e. by considering $\alpha = \alpha_1 \in \mathbb{R}$ ($K = 1$) and $y = (x, \alpha_1)$. It consists in finding the sequence of points $\{y^0, y^1, \ldots, y^j, y^{j+1}, \ldots\}$ that satisfy the equilibrium condition

$$F(y) = 0. \tag{3}$$

The first point $y^0 = (x^0, \alpha_1^0)$ is obtained by fixing α_1^0 and computing x^0 by means of Newton–Raphson iterations.[1] The following points are computed by standard prediction–correction methods as described below.

3.1.1. Prediction

Assuming that a point y^j of the continuation sequence is known, a prediction of the next point is obtained with

$$\tilde{y}^{j+1} = y^j + h^j v^j, \tag{4}$$

where $h^j \in \mathbb{R}$ is the step-size, and $v^j \in \mathbb{R}^{N+1}$ is the normalized tangent vector to (3) at y^j (see Fig. 3). The initial tangent vector v^0 is computed from $J(y^0)\, v^0 = 0$, where $J = \partial F/\partial y$, including a restriction on the norm of v^0. To do this, the value of one of its components is fixed at 1, and the other components are computed by solving the resulting system [43,44]. Then, the obtained vector is normalized such that $\|v^0\| = 1$. The following vectors are computed analogously by solving the extended system

$$\begin{bmatrix} J(y^j) \\ (v^{j-1})^\intercal \end{bmatrix} v^j = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tag{5}$$

where v^{j-1} is the tangent vector at y^{j-1} (computed at the previous iteration) and the jacobian matrix J is computed by finite differences. The last row is added to satisfy the normalization $\langle v^{j-1}, v^j \rangle = 1$ and to preserve the orientation of the continuation branch.

3.1.2. Correction

Once the prediction $\tilde{y}^{j+1}$ is obtained, the next point of the sequence can be obtained, for example, with the *pseudo-arclength continuation* technique [43,45]. This approach searches for the point y^{j+1} on a direction given by the hyperplane orthogonal to the vector v^j (see Fig. 3) which is represented by the restriction

$$g^j(y) = \langle y - \tilde{y}^{j+1}, v^j \rangle = 0, \qquad g : \mathbb{R}^{N+1} \to \mathbb{R}. \tag{6}$$

Therefore, y^{j+1} is obtained by solving numerically the set of equations

$$\begin{cases} F(y) &= 0, \\ g^j(y) &= 0, \end{cases} \tag{7}$$

by means of the Newton-like procedure

$$y^{(i+1)} = y^{(i)} + \Delta y^{(i)}, \qquad i = \{0, 1, 2, \ldots\}. \tag{8}$$

The initial point is selected as $y^{(0)} = \tilde{y}^{j+1}$, and the displacement $\Delta y^{(i)} = (\Delta x^{(i)}, \Delta \alpha_1^{(i)})$ is obtained from the linearization of (7) by applying the Newton-Chord approach, i.e. without updating the jacobian matrix during the iterations.[2] Thus, the displacement is computed by solving

$$\begin{bmatrix} J(y^{(0)}) \\ g_y^j(y^{(0)}) \end{bmatrix} \Delta y^{(i)} = - \begin{bmatrix} F(y^{(i)}) \\ g^j(y^{(i)}) \end{bmatrix}, \tag{9}$$

where $g_y^j = \partial g^j/\partial y$ and $g_y^j(y^{(0)}) = v^{j\intercal}$ is the tangent vector computed during the prediction step.

3.1.3. Step-size control

The step-size control is based on the convergence behavior when solving numerically (7). It is increased when the number of iterations is relatively low (to speed-up the continuation), it is reduced when they exceed a given upper bound (to prevent convergence issues) and it is not modified when they are between these limits. On the contrary, when the algorithm does not converge after a sufficiently large number of iterations, the step-size is reduced and the procedure is repeated. If the process does not converge after a given number of attempts the continuation is aborted.

3.2. Detection and location of Hopf bifurcations

A necessary condition for the occurrence of a Hopf bifurcation is that a pair of complex conjugate eigenvalues (one mode) of the linearization matrix $A = \partial f/\partial x$, evaluated at a given operation condition, has zero real part. Therefore, in order to detect and locate a Hopf bifurcation, a scalar test function ψ, which is zero at the bifurcation condition, must be defined. Then, assuming that a convenient test function is defined, a Hopf bifurcation is detected between two consecutive points y^j and y^{j+1} of the equilibrium sequence, when the sign of this function evaluated at these points changes, i.e. when

$$\psi(y^j)\,\psi(y^{j+1}) < 0. \tag{10}$$

Finally, the actual Hopf bifurcation point satisfying

$$\begin{cases} F(y) = 0, \\ \psi(y) = 0, \end{cases} \tag{11}$$

is computed (located) by means of an iterative procedure.

Solving (11) by means of Newton's like methods presents some challenges since, in general, it is not possible to express the derivatives of the test function in explicit form. Therefore, the jacobian matrix associated to the left-hand side of (11) must be computed through numerical differentiation methods which usually involves a relatively large computational load. A useful approach that avoids computing the jacobian matrix, providing a reasonable balance between the computational cost and the velocity of convergence is the so called Broyden update method [46,47]. The method replaces the jacobian matrix, which is used to obtain the displacement $\Delta y^{(i)}$, by a matrix that is derived from considering the secant equation corresponding to the last two points of the iterations, namely $y^{(i-1)}$ and $y^{(i)}$. Thus, the actual Hopf bifurcation point is obtained through the iterative procedure (8) with the displacement given by

$$\Delta y^{(i)} = -H^{(i)} \phi^{(i)}, \tag{12}$$

[1] As is usual in power systems stability studies, this step starts from a desired power flow solution which defines the border conditions to initialize the dynamic models (DFIG, transmission lines, etc.).

[2] This strategy is used in order to reduce the computational load. If convergence is not achieved after a predefined number of iterations the standard NR approach may be used [39].

where $\phi^{(i)} = \left(F\left(y^{(i)}\right), \psi\left(y^{(i)}\right)\right)$ and $H \in \mathbb{R}^{(N+1)\times(N+1)}$ is updated through

$$H^{(i)} = H^{(i-1)} - \left(\Delta y^{(i-1)} + H^{(i-1)}\Delta\phi^{(i-1)}\right)\frac{\left(\Delta y^{(i-1)}\right)^{\intercal} H^{(i-1)}}{\left(\Delta y^{(i-1)}\right)^{\intercal} H^{(i-1)}\Delta\phi^{(i-1)}}, \tag{13}$$

with $\Delta\phi^{(i-1)} = \phi^{(i)} - \phi^{(i-1)}$. The initial guess is selected as $y^{(0)} = y^{j+1}$ and the procedure for selecting $H^{(0)}$ is described in Appendix A.1.

3.3. Continuation of Hopf bifurcations

The continuation of Hopf bifurcations requires the variation of two parameters ($K = 2$), i.e. considering that now $\alpha = (\alpha_1, \alpha_2) \in \mathbb{R}^2$. Defining $y_H = (x, \alpha_1, \alpha_2) \in \mathbb{R}^{N+2}$, the continuation problem results

$$\begin{cases} F(y_H) = 0, \\ \psi(y_H) = 0. \end{cases} \tag{14}$$

The first point of the sequence that satisfies (14), namely $y_H^0 = (x^0, \alpha_1^0, \alpha_2^0)$, is obtained with the values x^0 and α_1^0 computed with the location procedure described in Section 3.2 and α_2^0 is the value of parameter α_2 that was considered fixed during the equilibrium continuation.

3.3.1. Prediction

The prediction $\tilde{y}_H^{j+1}$ is obtained through

$$\tilde{y}_H^{j+1} = y_H^j + h^j v_H^j, \tag{15}$$

with the direction v_H given by the secant vector[3] at the last two points of the Hopf bifurcation sequence, namely y_H^j and y_H^{j-1}, i.e. by means of

$$v_H^j = \frac{y_H^j - y_H^{j-1}}{\|y_H^j - y_H^{j-1}\|}. \tag{16}$$

To compute the initial vector, in this case v_H^1, a point y_H^1 sufficiently close to y_H^0 on the Hopf continuation curve is obtained by perturbing α_2^0 slightly to α_2^1 and using the location process described previously, i.e. with (8) and (12)–(13), to obtain $y^1 = (x^1, \alpha_1^1)$ and thus $y_H^1 = (x^1, \alpha_1^1, \alpha_2^1)$.

3.3.2. Correction

Once a prediction $\tilde{y}_H^{j+1}$ is obtained, the point y_H^{j+1} satisfying (14) is obtained following the pseudo-arclength approach, i.e. by solving the extended system

$$\begin{cases} F(y_H) = 0, \\ \psi(y_H) = 0, \\ g^j(y_H) = 0, \end{cases} \tag{17}$$

with

$$g^j(y_H) = \langle y_H - \tilde{y}_H^{j+1}, v_H^j \rangle, \qquad g : \mathbb{R}^{N+2} \to \mathbb{R}. \tag{18}$$

The solution is obtained by the iterative process

$$y_H^{(i+1)} = y_H^{(i)} + \Delta y_H^{(i)}, \qquad i = \{0, 1, 2, \ldots\}. \tag{19}$$

The displacement is computed using the Broyden method

$$\Delta y_H^{(i)} = -H_H^{(i)}\phi_H^{(i)} \tag{20}$$

where $\phi_H^{(i)} = \left(F\left(y_H^{(i)}\right), \psi\left(y_H^{(i)}\right), g\left(y_H^{(i)}\right)\right)$ and $H_H \in \mathbb{R}^{(N+2)\times(N+2)}$ is updated through

$$H_H^{(i)} = H_H^{(i-1)} - \left(\Delta y_H^{(i-1)} + H_H^{(i-1)}\Delta\phi_H^{(i-1)}\right)\frac{\Delta y_H^{(i-1)\intercal} H_H^{(i-1)}}{\Delta y_H^{(i-1)\intercal} H_H^{(i-1)}\Delta\phi_H^{(i-1)}}, \tag{21}$$

with $\Delta\phi_H^{(i-1)} = \phi_H^{(i)} - \phi_H^{(i-1)}$. The initial guess is selected as $y_H^{(0)} = \tilde{y}_H^{j+1}$ and the procedure for selecting $H_H^{(0)}$ is described in Appendix A.2.

[3] This approach is used instead of the tangent approach, to avoid computing derivatives of the test function.

3.3.3. Step-size control

The step-size control strategy when solving numerically (17) follows the same guidelines described in Section 3.1.3.

4. Hopf bifurcation test functions

A straightforward implementation of the test function is given by $\psi(y) = \prod_{i>j}(\lambda_i(y)+\lambda_j(y))$, where λ_i and λ_j, are any of the N eigenvalues of the linearization matrix A. Clearly, this function vanishes at the Hopf bifurcation condition, i.e. when some $\lambda_i(y) = j\omega_H$ and $\lambda_j(y) = -j\omega_H$ (the mode is on the imaginary axis). However, this expression requires the computation of all the eigenvalues of the linearization matrix A. Usually, this task is avoided using equivalent but more complex expressions such as $\psi(y) = \det(2A(y) \odot I_N)$, where $\odot$ denotes the bialternate product and $I_N \in \mathbb{R}^{N\times N}$ is the identity matrix (see [43]), or $\psi(y) = \det\left(A(y) \otimes I_N + I_N \otimes A(y)\right)$, where $\otimes$ is the tensor product (see [48]). Nevertheless, in both cases it is necessary to deal with determinants of square matrices of order $N(N-1)/2$, in the first case, and N^2 in the second case. For example, the model of the relatively simple system of Fig. 1 has $N = 690$ states and the order of the corresponding matrices results in 237,705 and 476,100, respectively. Therefore, these test functions are not applicable for the continuation of Hopf bifurcations in practical power system models. To overcome this limitation, a simple test function is proposed.

4.1. Proposed test function

The stability boundary associated to the SSO phenomenon is defined by the locus of the Hopf bifurcation exhibited by the subsynchronous mode. Since this mode is physically related to the supersynchronous mode, the test function can be defined focusing the attention only on these modes. Therefore, denoting the eigenvalues composing the subsynchronous mode as $\lambda_{sb\pm}(y) = \sigma_{sb} \pm i\omega_{sb}$, and the ones of the supersynchronous mode as $\lambda_{sp\pm}(y) = \sigma_{sp} \pm i\omega_{sp}$, the proposed test function is[4]

$$\psi(y) = \frac{\sigma_{sb}\,\sigma_{sp}}{|\sigma_{sb}| + |\sigma_{sp}|}. \tag{22}$$

Clearly, this function vanishes at the Hopf bifurcation condition, i.e. when $\sigma_{sb} = 0$, and hence $\lambda_{sb\pm} = \pm i\omega_{sb}$. Notice that $\psi = 0$ also when the real part of the supersynchronous mode is $\sigma_{sp} = 0$. Besides, this situation is not usual, both cases can be easily distinguished by the frequency of the modes. In addition, the denominator is included to smooth the function around $\psi = 0$, promoting the convergence of the continuation routine.

4.2. Computation of $\lambda_{sb\pm}$ and $\lambda_{sp\pm}$

The proposed test function (22) does not present the drawback with the order of the involved matrices that have the ones based on the bialternate and the tensor products described before. Nevertheless, (22) requires to compute the subsynchronous and supersynchronous modes every time the test function ψ must be evaluated. This task can be carried out using the Selective Modal Analysis (SMA) method [40,41], which is an iterative procedure that reproduces a portion of the eigenstructure of the jacobian matrix corresponding to the full power system, avoiding the computation of all the eigenvalues and eigenvectors. In [42], the SMA method has been adapted to provide a reduced matrix $A_R \in \mathbb{R}^{4\times4}$, i.e. with the minimum dimension, that reproduces the eigenstructure of A associated to the subsynchronous and supersynchronous modes involved in the SSO phenomenon. The adapted method consists in a two-phase systematic procedure. In the first phase, the attention is focused on the so called electrical subsystem, obtained from matrix A by considering only the electrical variables (bus voltages,

[4] The explicit dependence of σ and ω on y is omitted for simplicity.

line currents, WF equivalent internal network voltage and current, and equivalent DFIG stator currents). The SMA approach is applied to this subsystem and a state transformation matrix that decouples the subsynchronous and supersynchornous modes from the high frequency modes associated with the network variables is derived. This decoupling between the electrical variables improves the convergence conditions of the SMA iterative method when the entire system is considered. Then, in the second phase, the full matrix with the transformed electrical variables is considered and the SMA algorithm successfully provides a reduced system matrix with the minimum dimension, i.e. only four state variables, reproducing the super and subsynchronous modes of interest. The adapted SMA method is described in detail in [42] and the main features are condensed in Appendix A.3.

4.3. Hopf bifurcation detection and continuation

The methodology for the detection and continuation of the Hopf bifurcation associated to the SSO phenomenon follows the general framework described in Sections 3.1–3.3 which is enabled by the proposed SMA based test function ψ given by (22). The procedure is summarized in the flowchart of Fig. 4. It requires the execution of the adapted SMA method to obtain σ_{sb} and σ_{sp} every time the test function ψ is evaluated, i.e. in the Hopf bifurcation detection and location routines, and in the correction step of the Hopf bifurcation continuation routine (these steps are highlighted in gray in the flowchart). In particular, the location and correction routines avoid the recursive computation of the jacobian matrices associated to the NR method by using the Broyden's approach.

5. Example

In order to evaluate the proposed methodology, the approach is used to detect and continue the stability boundary associated to the SSO phenomenon on the power system of Fig. 1. In this example, all the WF are operating at a relatively low wind condition, with a wind speed $v_w = 5.5$ m/s, and the rated power indicated in Table 1.

5.1. Equilibrium continuation and detection of the critical condition (Hopf bifurcation)

The continuation procedure, condensed in the flowchart of Fig. 4, begins with the equilibrium continuation and the Hopf bifurcation detection. In this case, the compensation level μ is selected as the continuation parameter ($\alpha_1 = \mu$) with an arbitrarily chosen initial value $\mu^0 = 0.6$. Then, the equilibrium point x^0 for $\mu^0 = 0.6$ is computed by standard NR iterations and the initial point $y^0 = (x^0, \mu^0)$ of the equilibrium continuation is obtained. The resulting point is depicted in Fig. 5 by an empty circle. For convenience, the result is shown in the parameter plane S_{PM1}–μ, where S_{PM1} is the power of the cluster PM1 which is left fixed at the nominal value (4.2 p.u. or 420 MW) during the equilibrium continuation.

The following points of the equilibrium sequence are obtained through the prediction–correction algorithm. In Fig. 5, the predictions $(\tilde{y}^1, \tilde{y}^2, \tilde{y}^3)$ and the corrections (y^1, y^2, y^3) are denoted by crosses and empty circles, respectively. Once the correction at a step j is finished, i.e. when y^{j+1} is obtained, the procedure runs the detection routine. Towards this end, the test function $\psi(y^{j+1})$ is evaluated using the subsynchronous and supersynchronous modes ($\lambda_{sb\pm}$ and $\lambda_{sp\pm}$) computed with the adapted SMA iterative method briefly described in Appendix A.3. The procedure is carried out selecting as relevant variables (x_{Er}) both components of the voltage of the compensation capacitor bank (between buses 10 and 11) and those of the current of the compensated line (between buses 1008 and 11). The computed values of the subsynchronous mode $\lambda_{sb\pm}$ at the equilibria y^0 to y^3 are shown in Fig. 6 (empty circles).

The performance of the equilibrium continuation and the Hopf bifurcation detection routines is summarized in Table 2. It shows, on one hand, the step-size h used in the prediction stage and the number of corrections required to obtain each equilibrium point of the sequence (for the initial point y^0 the value corresponds to the number of NR iterations). On the other hand, it shows the number of iterations required at the first and second phases of the adapted SMA method, the obtained subsynchronous and supersynchronous modes and the corresponding value of the test function. The elapsed time at each continuation step (including equilibrium continuation and Hopf bifurcation detection) is shown at the last column. The procedure was executed on a desktop computer with an Intel Core i5-4670K 3.4 GHz processor and 16 GB of RAM.

A Hopf bifurcation is detected when the sign of the test function changes between two consecutive points of the sequence, i.e. when the subsynchronous mode crosses the imaginary axis. This situation occurs with the equilibrium points y^2 ($\mu = 0.695$) and y^3 ($\mu = 0.746$) as is clearly seen in Table 2 and in Fig. 6. Then, the algorithm triggers the location procedure described in Section 3.2, and the actual bifurcation point y_H^0, denoted by a black dot on the imaginary axis in Fig. 6, is located for $\mu = 0.703$.

5.2. Stability boundary (Hopf bifurcation) continuation

In order to continue the Hopf bifurcation that defines the stability boundary associated to the SSO problem, a second parameter must be varied. Initially, any parameter of interest can be chosen. In this case, the power available at cluster PM1 is selected. This selection exposes the potential of the continuation tool when dealing with non trivial operating conditions, such as those arising when the number of wind turbines in operation at a given cluster changes, for example, due to the tripping of turbines. For simplicity, this scenario is represented by varying continuously the available power at a given cluster as $S_x = K_x S_x^*$, where x is replaced by PM1, PM2, CR or SCN denoting the corresponding cluster, and $0 < K_x \leq 1$ is the fraction of the corresponding rated power at that cluster, namely S_x^*. In addition, following [12], the equivalent network and 0.69/33 kV transformer impedances are scaled by the same factor as $Z_x = Z_x^*/K_x$, where Z_x^* is the corresponding rated equivalent impedance.

Therefore, selecting S_{PM1} as a second parameter ($\alpha_2 = S_{PM1}$) the prediction–correction method is used to compute the Hopf bifurcation locus as parameters μ and S_{PM1} are varied simultaneously. The predictions $\tilde{y}_H^1$ to $\tilde{y}_H^3$ are denoted in Fig. 5 by crosses, and the resulting Hopf bifurcation points y_H^1 to y_H^3 are denoted by black dots (in addition to the initial point y_H^0 computed previously). The corresponding subsynchronous mode (moving along the imaginary axis) is shown in Fig. 6. The performance of the Hopf bifurcation location and continuation routines is shown in Table 3. The results for the Hopf bifurcation location routine is included in the first row and those associated to the Hopf bifurcation continuation are shown in the last three rows. Table 3 shows the step size h, the number of iterations of the Broyden's method (column labeled Corr.), the number of iterations required by the first and second phases of the adapted SMA method (which is repeated for each Broyden update), the obtained subsynchronous and supersynchronous modes, the value of the test function evaluated at each point of the continuation, and the total elapsed time. In particular, it is worth noticing that the computed subsynchronous mode $\lambda_{sb\pm}$ has zero real part in all the cases, validating the effectiveness of the implemented method in detecting the critical condition. The tolerances used for the Broyden's method are 1×10^{-4} for both, the norm of the extended function (17) and the displacement Δy_H. In addition, the ones for phases 1 and 2 of the SMA method are $\epsilon_1 = 10^{-1}$ (see (A.14)) and $\epsilon_2 = 10^{-4}$ (see (A.19)), respectively.

The stability boundary in the plane $S_{PM1} - \mu$ is obtained by extending the Hopf bifurcation continuation sequence up to $\mu = 1$. The result is depicted in Fig. 7, where the black dots denote the points

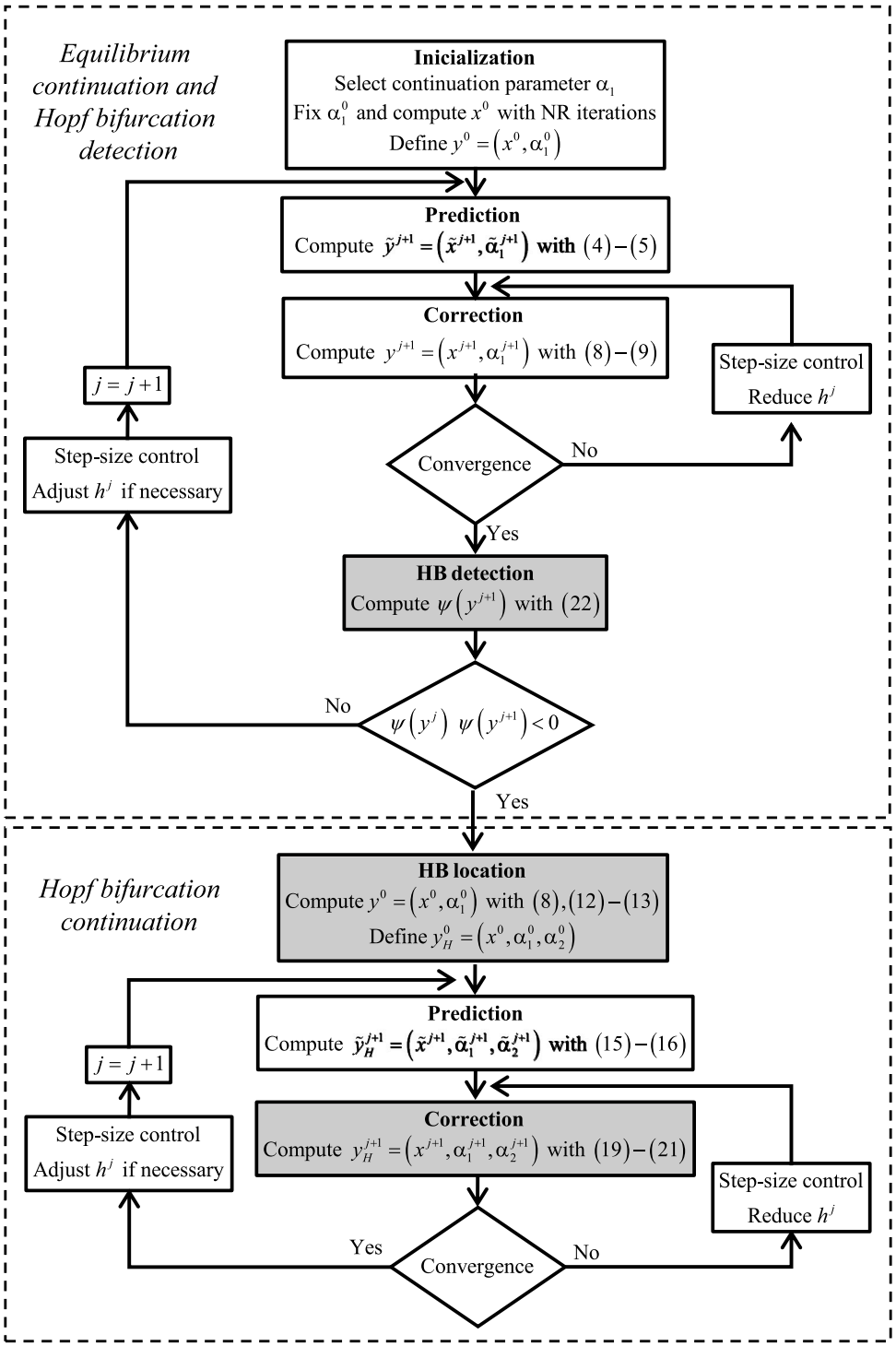


Fig. 4. Flowchart representing the detection and continuation procedure. The steps that require computing σ_{sb} and σ_{sp} using the adapted SMA method are highlighted in gray.

Table 2
Performance of the equilibrium point continuation and Hopf bifurcation detection routines.

Equilibrium continuation			Hopf bifurcation detection				T [ms]
Point	h	Corrections	SMA[a]	$\lambda_{sb\pm}$	$\lambda_{sp\pm}$	$\psi(y^j)$	
y^0	–	3	3/6	$-1.3662 \pm j206.61$	$-13.858 \pm j419.98$	1.24	522
y^1	0.05	2	3/6	$-0.7478 \pm j202.44$	$-13.938 \pm j424.19$	0.71	632
y^2	0.05	2	3/6	$-0.1235 \pm j198.43$	$-14.011 \pm j428.24$	0.12	631
y^3	0.05	2	3/6	$0.5073 \pm j194.57$	$-14.079 \pm j432.14$	−0.49	631

[a]Number of iterations of the two-phases adapted SMA methods (phase 1/phase 2).

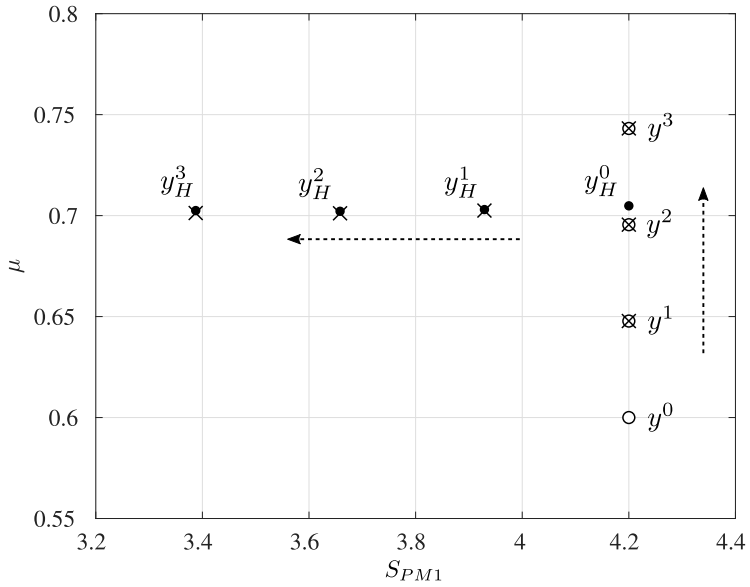


Fig. 5. Results of equilibrium point initialization (empty circle y^0) and continuation (empty circles y^{j+1}), Hopf bifurcation location (black dot y_H^0) and continuation (black dots y_H^{j+1}). Predictions are denoted by crosses.

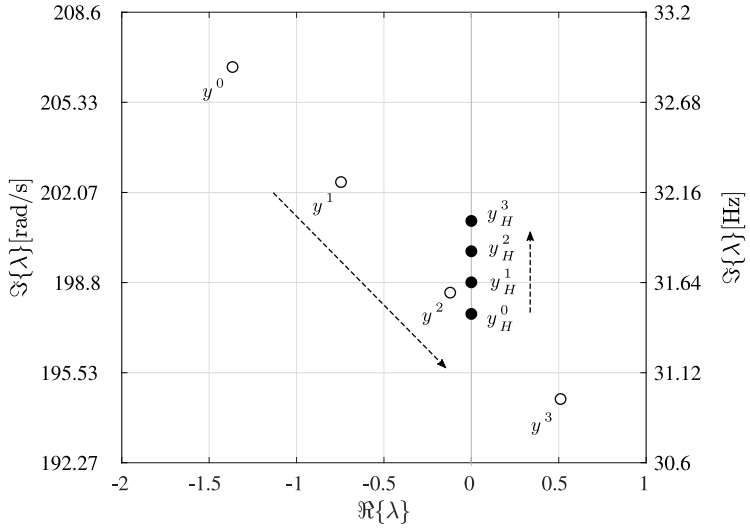


Fig. 6. Subsynchronous mode (positive imaginary part) obtained with the adapted SMA method for detecting Hopf bifurcations during the equilibrium continuation (empty circles) and for the Hopf bifurcation continuation (black dots).

Table 3
Performance of the Hopf bifurcation location and continuation routines.

Point	h	Corr.	SMA	$\lambda_{sb\pm}$	$\lambda_{sp\pm}$	$\psi\left(y^j\right)$	T [ms]
y_H^0	–	3	3/6	$0.0000 \pm j197.66$	$-14.025 \pm j429.01$	-0.02×10^{-6}	923
y_H^1	1	2	3/6	$0.0000 \pm j198.80$	$-14.034 \pm j427.89$	-1.05×10^{-6}	1767
y_H^2	1	2	3/6	$0.0000 \pm j199.93$	$-14.043 \pm j426.79$	-2.25×10^{-6}	1757
y_H^3	1	2	3/6	$0.0000 \pm j201.03$	$-14.053 \pm j425.72$	-2.76×10^{-6}	1761

y_H computed with the algorithm, and those labeled as y_H^0 to y_H^3 are the ones described previously and included in Table 3 and Fig. 5. This stability boundary delimits stable operating conditions on the left side of the curve and unstable ones on the right side. Therefore, for compensation levels lower than 70% ($\mu < 0.7$), the system is stable for the full range of powers at the cluster PM1 ($0 \leq S_{PM1} \leq 4.2$). On the other hand, compensation levels larger than 70% reduce the stable operating region. Thus, for example, for $\mu = 0.75$, the power at PM1 is limited to approximately 1.845 p.u. (44% of the nominal value) and the system is unstable for larger values of S_{PM1} (see the dashed lines in Fig. 7). To illustrate the behavior of the SSO corresponding to the stable and unstable regions, time domain simulations with the cluster PM1 at the nominal installed power (vertical line at $S_{PM1} = 4.2$) were performed. The results, included in Fig. 8, show the evolution of the active power of the cluster PM1 after introducing a perturbation at bus 12. The simulation in blue (at the top) corresponds to the scenario with

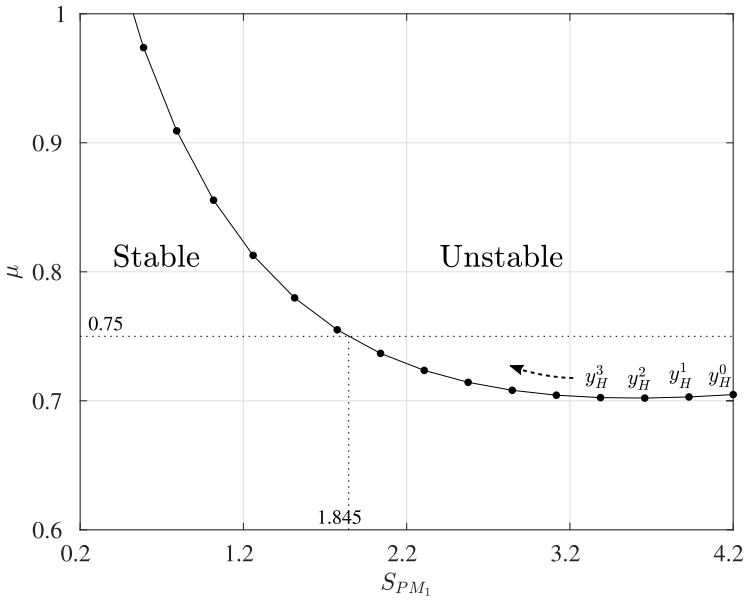


Fig. 7. Stability boundary (Hopf bifurcation curve) in the plane given by the power of cluster PM1 (S_{PM1}) and the compensation level (μ). The nominal power of PM1 is 4.2 p.u. for a base power of 100 MVA. The clusters PM2, CR and SCN are at their nominal power.

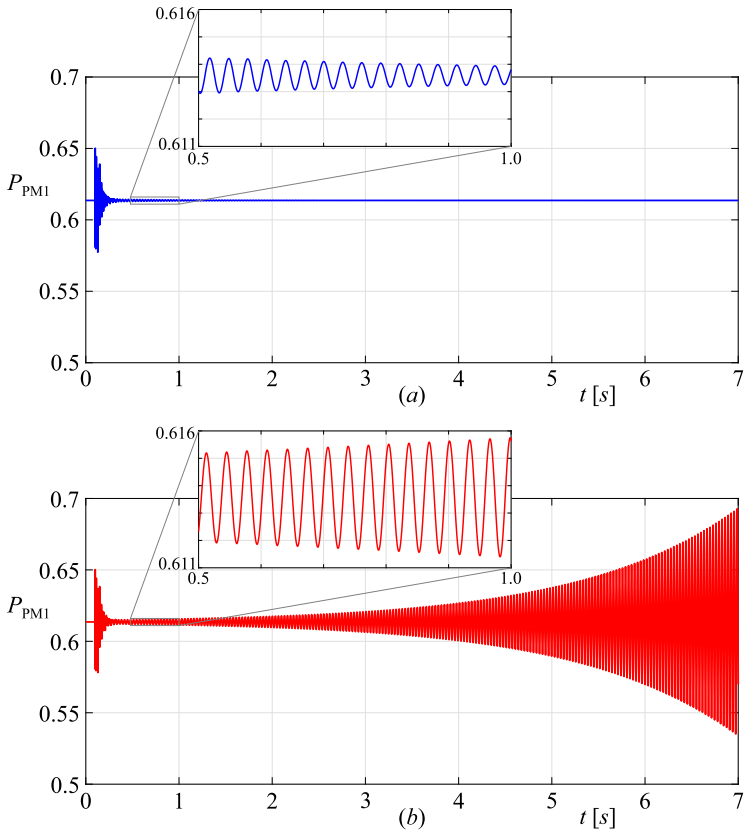


Fig. 8. Simulations corresponding to the active power of the cluster PM1 (at nominal installed power) for a stable scenario with $\mu = 0.6$ (blue) and an unstable scenario with $\mu = 0.75$ (red).

$\mu = 0.6$, and the attenuation of the amplitude of the SSO (≃33 Hz in the *dq* frame) reveals that the scenario is stable in accordance to Fig. 7 (the point with $\mu = 0.6$ and $S_{PM1} = 4.2$ is located in the stable region). On the other hand, in the scenario with $\mu = 0.75$ (red simulation at the bottom), the amplitude of the SSO (≃30 Hz) grows and corresponds to an unstable case in agreement with Fig. 7 (the point with $\mu = 0.75$ and $S_{PM1} = 4.2$ is located in the unstable region).

5.3. Multiparametric stability analysis

The analysis of scenarios different from the one considered in Fig. 7 can be performed by changing the power of clusters CR, SCN or PM2 and repeating the continuation procedure. For example, Fig. 9 shows the stability boundaries obtained for different values of the power at the cluster CR, namely the nominal case $S_{CR} = 1.46$ (black), $S_{CR} =$

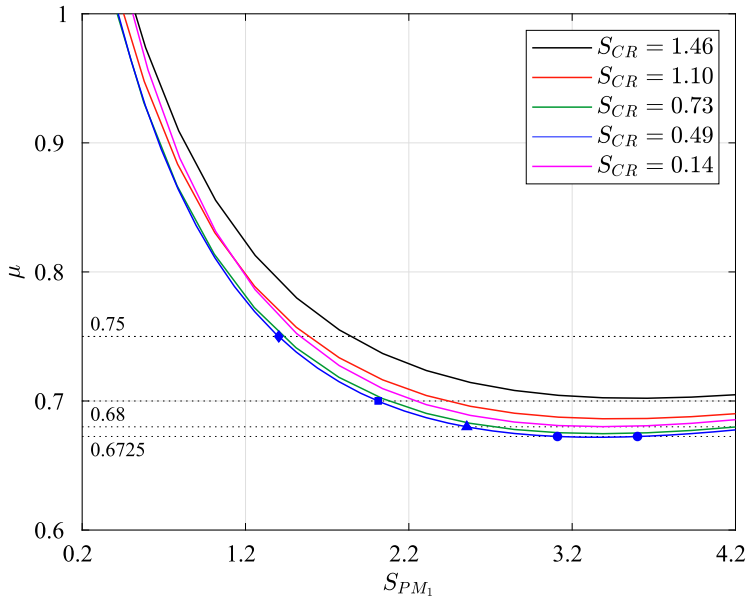


Fig. 9. Stability boundary (Hopf bifurcation curve) in the plane $S_{PM1} - \mu$ for different values of the available power at cluster CR. The nominal power of clusters PM1 and CR are 4.2 p.u. and 1.46 p.u., respectively, for a base power of 100 MVA. The clusters PM2 and SCN are at their nominal power.

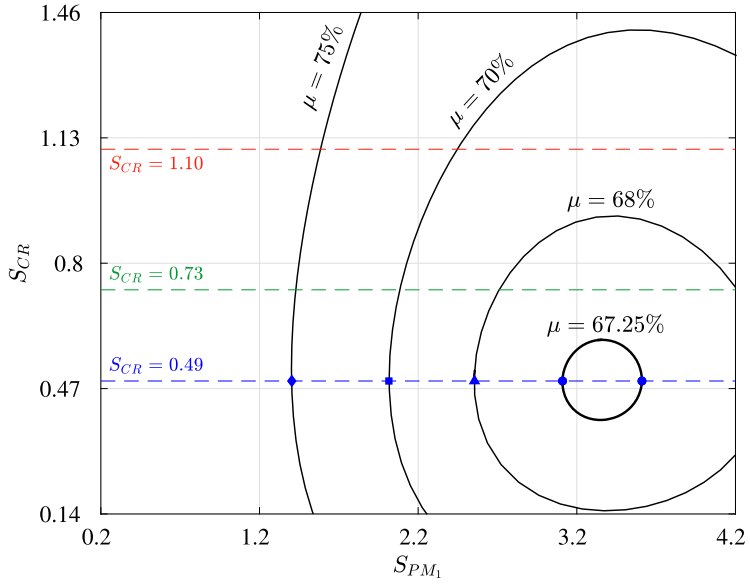


Fig. 10. Stability boundary (Hopf bifurcation curve) in the plane S_{PM1}–S_{CR} for different values of the compensation level.

1.10 (red), $S_{CR} = 0.79$ (green), $S_{CR} = 0.49$ (blue) and $S_{CR} = 0.14$ (magenta). As described previously, the equilibrium point is stable for operating conditions on the left side of the curves and unstable on the right side. These results reveal that a reduction of the power at CR from the nominal value $S_{CR} = 1.46$ to $S_{CR} = 0.49$ moves the stability boundary downwards, limiting even more the stable operating region. A reduction of S_{CR} below 0.49 moves the curve upwards, in the direction of the nominal case, ceasing the deterioration of the stability region as is depicted by the curve corresponding to $S_{CR} = 0.14$ (magenta). Therefore, the blue curve for $S_{CR} = 0.49$ (1/3 of the nominal power at CR), represents a worst case scenario when PM2 and SCN are at their nominal powers.

The effect of the available power at the clusters PM1 and CR on the stability boundary is better exposed by considering S_{PM1} and S_{CR} as continuation parameters. The results for five different values of the compensation level μ (with PM2 and SCN at their nominal values) are shown in Fig. 10. To support the interpretation of the results, the values of S_{CR} corresponding to the boundaries derived in Fig. 9 are identified (horizontal dashed lines). In addition, the values of μ corresponding to the curves in Fig. 10 are also indicated in Fig. 9. Selecting the worst case scenario, corresponding to $S_{CR} = 0.49$, the critical points for the different compensation values are identified by blue markers in both figures. For $\mu = 75\%$ (blue diamond), both figures reveal that the power at PM1 is severely restricted to approximately 1.4 p.u. (30% of the nominal value), and it must be lower than this limit in order to be stable for any value of the power at CR. This means that the stable operating condition is on the left of the outermost boundary (blue curve) in Fig. 9 and that a vertical line in Fig. 10 for S_{PM1} below 1.4 p.u. does not intersect the stability boundary corresponding to $\mu = 0.75$. For lower compensation levels, the limitation on S_{PM1} is reduced (the critical points move towards the right). In particular, for $\mu = 0.6725$, the boundary is a relatively small closed curve or bubble, enclosing the unstable operating conditions. This particular shape of the stability boundary explains behaviors related to wind turbine tripping observed

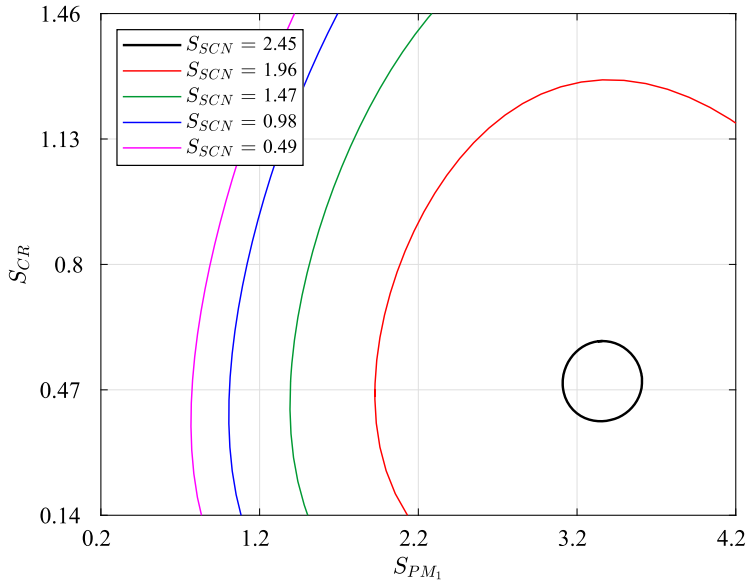


Fig. 11. Stability boundary in the plane S_{PM1}–S_{CR} for different values of S_{SCN} for a fixed compensation level $\mu = 0.6725$ and PM2 at nominal power.

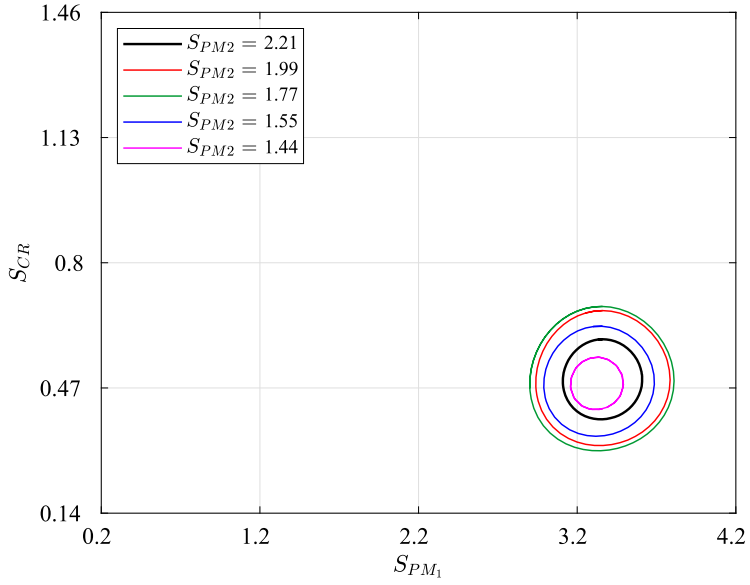


Fig. 12. Stability boundary in the plane S_{PM1}–S_{CR} for different values of S_{PM2} for a fixed compensation level $\mu = 0.6725$ and SCN at nominal power.

in practical systems with multiple WF [4]. A contingency can drive the system to an operation condition inside the bubble, triggering a SSO event. Then, the subsequent tripping of wind turbines reduces the power of the WF, the operating condition moves outside the bubble and the system regains stability [12]. Therefore, operating conditions located near the point $S_{PM1} \simeq 3.4$ and $S_{CR} \simeq 0.49$ are prone to exhibit undamped SSO when the clusters SCN and PM2 are at their nominal powers. This case reveals one of the main capabilities of the proposed methodology, which consists in the detection of worst case conditions in scenarios with multiple wind farms. This information can be used later to focus the attention on these conditions when conducting complementary studies by using practical methods, such as impedance modeling and EMT simulations.

To illustrate the capability of the method in detecting non trivial scenarios, the analysis is extended by considering the effect of the power at the clusters SCN and PM2 on the stability boundary. Towards this end, the compensation level is fixed at $\mu = 0.6725$, which corresponds to the smaller bubble in Fig. 10, and the proposed continuation approach is used to obtain the boundary for different values of the power at SCN (keeping PM2 at the nominal value) and, then, for different values of the power at PM2 (keeping SCN at the nominal value). The effect of reducing the power at SCN is shown in Fig. 11, where the growth of the bubbles enclosing the unstable region, is evident. For example, for $S_{SCN} = 0.49$ (magenta) the power at PM1 is significantly restricted to the left of the curve revealing a significant deterioration of the system stability. On the other hand, the effect of the power at the cluster PM2 on the stability boundary is shown in Fig. 12. As is appreciated, the black bubble corresponding to the nominal value $S_{PM2} = 2.21$ (which is the reference boundary for $\mu = 0.6725$ in Fig. 10) grows to the green curve when the power is reduced to $S_{PM2} = 1.77$ (80% of the nominal value), enlarging moderately the unstable region. A further reduction below $S_{PM2} = 1.77$ shrinks the bubbles (see, for example, the magenta curve for $S_{PM2} = 1.44$) until they collapse and disappear for $S_{PM2} < 1.44$.

Finally, the analysis performed in Fig. 10 with SCN and PM2 at their nominal powers can be repeated for a worse condition with $S_{SCN} =$

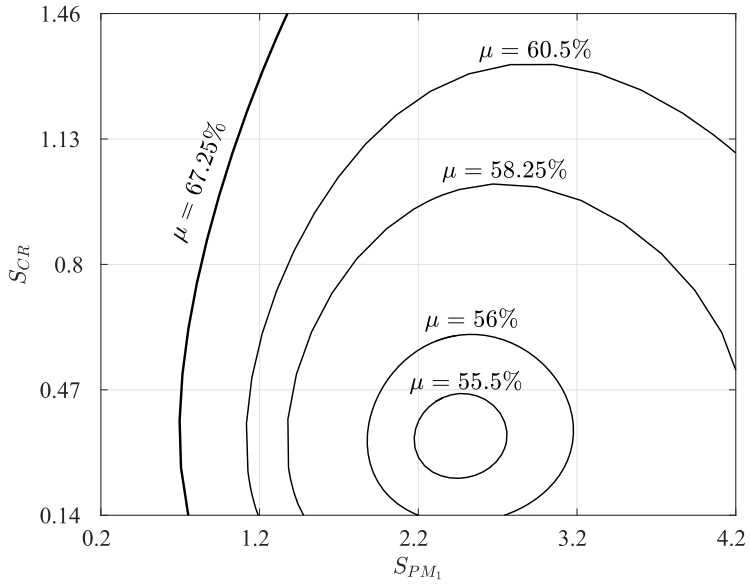


Fig. 13. Stability boundary (Hopf bifurcation curve) in the plane S_{PM1}–S_{CR} for different values of the compensation level with $S_{SCN} = 0.49$ and $S_{PM2} = 1.77$.

0.49 and $S_{PM2} = 1.77$. The result is shown in Fig. 13 where the small black bubble for $\mu = 0.6725$ in Fig. 10 grew significantly limiting the power at PM1 to approximately 0.7 p.u. The stability boundary moves to the right and shrinks when the compensation level is reduced and, for example, a relatively small bubble is obtained for $\mu = 0.55$. A reduction of the compensation level below $\mu = 0.55$ precludes the emergence of undamped SSO. Fig. 13 also reveals the convexity of the stability boundary, since the curves shrink for decreasing values of the compensation level until they collapse for a sufficiently small value of μ, suggesting that for $\mu \simeq 0.55$, the worst condition occurs for $S_{PM1} \simeq 2.4$ and $S_{CR} \simeq 0.35$ (with $S_{SCN} = 0.49$ and $S_{PM2} = 1.77$).

6. Conclusions

In this paper, the detection and continuation of the Hopf bifurcation defining the stability boundary associated to SSO in DFIG-based WF was addressed. A test function based only on the real part of the subsynchronous and supersynchronous modes was proposed. The required modes are obtained with an adapted version of the SMA iterative method developed previously. Then, the standard prediction–correction numerical algorithms used for the detection and continuation of the Hopf bifurcation were adapted to include the proposed test function along with the SMA based computation of the modes. This approach overcomes the main limitation of the test function implemented in standard continuation packages for studying SSO in large scale power system models. The methodology was illustrated considering a system with 690 state variables, composed by 11 DFIG-based WF represented by 22 aggregated models. The obtained results reveal the potential of the numerical continuation tools to go further in the analysis of the stability boundary associated to the SSO phenomenon in large scale power systems.

CRediT authorship contribution statement

Nicolás E. Costa: Methodology, Conceptualization, Investigation, Software, Validation, Visualization, Writing – review & editing. **Gustavo Revel:** Methodology, Conceptualization, Investigation, Software, Validation, Visualization, Writing – review & editing. **Diego M. Alonso:** Methodology, Conceptualization, Writing – original draft, Writing – review & editing, Supervision. **Roberto D. Fernández:** Methodology, Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This work was supported by UNS, Argentina (PGI 24/K087) and CONICET, Argentina (PIP 1122020010 2076CO).

Appendix

A.1. Initialization of H for the location of Hopf bifurcations

Broyden's update method is based on replacing the jacobian matrix used in Newton's-like methods by a matrix B that satisfies the secant equation. In particular, the initial matrix $B^{(0)}$ used in the location of the Hopf bifurcation is forced to satisfy the secant equation at the last two points of the equilibrium continuation, y^j and y^{j+1}, this is

$$B^{(0)} \Delta y^j = \Delta \phi^j. \tag{A.1}$$

Then, giving a convenient structure to matrix B, (A.1) results

$$\begin{bmatrix} B^{(0)}_{F_x} & B^{(0)}_{F_{\alpha_1}} \\ B^{(0)}_{\psi_x} & B^{(0)}_{\psi_{\alpha_1}} \end{bmatrix} \begin{bmatrix} \Delta x^j \\ \Delta \alpha_1^j \end{bmatrix} = \begin{bmatrix} F\left(y^{j+1}\right) - F\left(y^j\right) \\ \psi\left(y^{j+1}\right) - \psi\left(y^j\right) \end{bmatrix}, \tag{A.2}$$

and the submatrices $B^{(0)}_{F_x}$, $B^{(0)}_{F_{\alpha_1}}$, $B^{(0)}_{\psi_x}$ and $B^{(0)}_{\psi_{\alpha_1}}$ must be selected in order to satisfy (A.2) or equivalently to guarantee that $(B^{(0)})^{-1} = H^{(0)}$ exists.

Since B plays the role of the jacobian, and considering that the linearization matrix $A\left(y^j\right)$ does not have eigenvalues at the origin, the following selection is straightforward

$$B^{(0)}_{F_x} = \left.\frac{\partial F}{\partial x}\right|_{(y^j)} = A\left(y^j\right). \tag{A.3}$$

Then, selecting for convenience

$$B^{(0)}_{\psi_x} = 0, \tag{A.4}$$

and considering that $F\left(y^{j+1}\right) = F\left(y^{j}\right) = 0$, the remaining submatrices are obtained from (A.2) and result

$$B^{(0)}_{F_{\alpha_1}} = -\frac{A\left(y^{j}\right)\Delta x^{j}}{\Delta\alpha_1^{j}}, \tag{A.5}$$

and

$$B^{(0)}_{\psi_{\alpha_1}} = \frac{\psi\left(y^{j+1}\right) - \psi\left(y^{j}\right)}{\Delta\alpha_1^{j}}. \tag{A.6}$$

Finally, $H^{(0)} = \left(B^{(0)}\right)^{-1}$ is obtained.

A.2. Initialization of H_H for the continuation of Hopf bifurcations

Proposing a structure for $B^{(0)}_H$ similar to the one used in (A.2), the corresponding secant equation results

$$\left[\begin{array}{ccc:c} B^{(0)}_{HF_x} & B^{(0)}_{HF_{\alpha_1}} & B^{(0)}_{HF_{\alpha_2}} & \\ B^{(0)}_{H\psi_x} & B^{(0)}_{H\psi_{\alpha_1}} & B^{(0)}_{H\psi_{\alpha_2}} & \\ \hdashline B^{(0)}_{Hg_x} & B^{(0)}_{Hg_{\alpha_1}} & B^{(0)}_{Hg_{\alpha_2}} & \end{array}\right]\left[\begin{array}{c}\Delta x^{j}\\ \Delta\alpha_1^{j}\\ \Delta\alpha_2^{j}\end{array}\right] = \left[\begin{array}{c} F\left(y_H^{j+1}\right) - F\left(y_H^{j}\right)\\ \psi\left(y_H^{j+1}\right) - \psi\left(y_H^{j}\right)\\ g\left(y_H^{j+1}\right) - g\left(y_H^{j}\right)\end{array}\right]. \tag{A.7}$$

The submatrices $B^{(0)}_{HF_x}$ and $B^{(0)}_{H\psi_x}$ are selected as before, i.e. $B^{(0)}_{HF_x} = A(y_H^{j+1})$ and $B^{(0)}_{H\psi_x} = 0$, and $B^{(0)}_{HF_{\alpha_1}}$ and $B^{(0)}_{H\psi_{\alpha_1}}$ are computed by means of finite differences perturbing α_1. Then, from the first two rows of (A.7) result

$$\left[\begin{array}{c} B^{(0)}_{HF_{\alpha_2}}\\ B^{(0)}_{H\psi_{\alpha_2}}\end{array}\right] = \frac{1}{\Delta\alpha_2^{j}}\left(\left[\begin{array}{c} F\left(y_H^{j+1}\right) - F\left(y_H^{j}\right)\\ \psi\left(y_H^{j+1}\right) - \psi\left(y_H^{j}\right)\end{array}\right] - \left[\begin{array}{cc} B^{(0)}_{HF_x} & B^{(0)}_{HF_{\alpha_1}}\\ B^{(0)}_{H\psi_x} & B^{(0)}_{H\psi_{\alpha_1}}\end{array}\right]\left[\begin{array}{c}\Delta x^{j}\\ \Delta\alpha_1^{j}\end{array}\right]\right). \tag{A.8}$$

Finally, the last row of $B^{(0)}_H$ is chosen as the secant vector v^0_H computed with (16), i.e.

$$\left[B^{(0)}_{Hg_x}\ B^{(0)}_{Hg_{\alpha_1}}\ B^{(0)}_{Hg_{\alpha_2}}\right] = v_H^{0\,\intercal}. \tag{A.9}$$

A.3. Adapted SMA method

Given the linearization matrix A at a given operation point, the subset of relevant variables $x_{Er} \in \mathbb{R}^4$ composed by the d–q components of the series compensation voltage and one of the line currents (e.g. the compensated line) is selected. The remaining line and collector currents, bus voltages, and stator currents are arranged in x_{Ez}, while the other variables (mechanical variables, controller and PLL states, etc.) are lumped in x_U. Then, the corresponding linear system is partitioned as

$$\left[\begin{array}{c}\dot{x}_{Er}\\ \dot{x}_{Ez}\\ \dot{x}_U\end{array}\right] = \left[\begin{array}{ccc} A_{Err} & A_{Erz} & A_{EUr}\\ A_{Ezr} & A_{Ezz} & A_{EUz}\\ A_{UEr} & A_{UEz} & A_U\end{array}\right]\left[\begin{array}{c}x_{Er}\\ x_{Ez}\\ x_U\end{array}\right], \tag{A.10}$$

and the adapted SMA routine has the phases described below.

Phase 1

A state transformation matrix K for the electrical subsystem is derived in order to reduce the participation of non relevant variables x_{Ez} on the subsynchronous and supersynchronous modes. This phase iteratively ($j = 1, 2, \ldots$) computes the matrices

$$\begin{aligned} A'^{(j)}_{Err} &= A'^{(j-1)}_{Err} + A_{Erz}\tilde{K}^{(j)},\\ A'^{(j)}_{Ezr} &= A'^{(j-1)}_{Ezr} - \tilde{K}^{(j)}A'^{(j-1)}_{Err} + \left(A'^{(j-1)}_{Ezz} - \tilde{K}^{(j)}A_{Erz}\right)\tilde{K}^{(j)},\\ A'^{(j)}_{Ezz} &= A'^{(j-1)}_{Ezz} - \tilde{K}^{(j)}A_{Erz}. \end{aligned} \tag{A.11}$$

where

$$\tilde{K}^{(j)} = \left[\kappa_1^{(j-1)}\hat{v}_{Er1}^{(j-1)} \cdots \kappa_4^{(j-1)}\hat{v}_{Er4}^{(j-1)}\right]\left[\hat{v}_{Er1}^{(j-1)} \cdots \hat{v}_{Er4}^{(j-1)}\right]^{-1}, \tag{A.12}$$

with

$$\kappa_i^{(j-1)} = \left(\hat{\lambda}_{Ei}^{(j-1)}I - A'^{(j-1)}_{Ezz}\right)^{-1}A'^{(j-1)}_{Ezr}. \tag{A.13}$$

The quantities $\hat{\lambda}_{Ei}$ and $\hat{v}_{Er1}^{(j-1)}$ are the ith eigenvalue and right eigenvector of $A_{Err}^{(j-1)}$. The algorithm finishes when the normalized distance

$$\delta_i^{(j)} = \left\|\hat{\lambda}_{Ei}^{(j)} - \hat{\lambda}_{Ei}^{(j-1)}\right\|\left\|\hat{\lambda}_{Ei}^{(j)}\right\|^{-1} < \epsilon_1, \quad i = \{1, \ldots, 4\}. \tag{A.14}$$

Phase 2

Computing $K = \sum \tilde{K}^{(j)}$ and defining the less relevant variables as $x_z = \left[x_{Ez} - Kx_{Er} \quad x_U\right]^{\intercal}$ the transformed whole system can be expressed as

$$\left[\begin{array}{c}\dot{x}_{Er}\\ \dot{x}_z\end{array}\right] = \left[\begin{array}{cc} A_{rr} & A_{rz}\\ A_{zr} & A_{zz}\end{array}\right]\left[\begin{array}{c}x_{Er}\\ x_z\end{array}\right], \tag{A.15}$$

where

$$\begin{aligned} A_{rr} &= A_{Err} + A_{Erz}K,\\ A_{rz} &= \left[A_{Erz} \quad A_{EUr}\right],\\ A_{zr} &= \left[\begin{array}{c} A_{Ezr} - KA_{Err} + \left(A_{Ezz} - KA_{Erz}\right)K\\ A_{UEr} + A_{UEz}K\end{array}\right],\\ A_{zz} &= \left[\begin{array}{cc} A_{Ezz} - KA_{Erz} & A_{EUz} - KA_{EUr}\\ A_{UEz} & A_U\end{array}\right]. \end{aligned} \tag{A.16}$$

Then

$$M^{(j)} = \left[H_1^{(j-1)}\hat{v}_{r1}^{(j-1)} \cdots H_4^{(j-1)}\hat{v}_{r4}^{(j-1)}\right]\left[\hat{v}_{r1}^{(j-1)} \cdots \hat{v}_{r4}^{(j-1)}\right]^{-1}, \tag{A.17}$$

with

$$H_i^{(j-1)} = A_{rz}\left(\hat{\lambda}_i^{(j-1)}I - A_{zz}^{(j-1)}\right)^{-1}A_{zr}^{(j-1)}. \tag{A.18}$$

The quantities $\lambda_i^{(j-1)}$ and $\hat{v}_{ri}^{(j-1)}$ for $j = \{2, 3, \ldots\}$ are obtained from $A_{rr} + M^{(j)}$. The initial values $\lambda_i^{(0)}$ and $\hat{v}_{ri}^{(0)}$ can be obtained from A_{rr} or from a previous knowledge of the system. The algorithm concludes when the normalized distance

$$\delta_i^{(j)} = \left\|\hat{\lambda}_i^{(j)} - \hat{\lambda}_i^{(j-1)}\right\|\left\|\hat{\lambda}_i^{(j)}\right\|^{-1} < \epsilon_2, \quad i = \{1, \ldots, 4\}. \tag{A.19}$$

Then, the eigenvalues of interest $\lambda_{sb\pm}$ and $\lambda_{sp\pm}$ correspond to $\hat{\lambda}_i$ obtained at the last iteration.

References

[1] Gross L. Sub-synchronous grid conditions: New event, new problem, and new solutions. In: 37th annual western protective relay conference spokane. Washington, USA; 2010, p. 1–19.

[2] Mulawarman A, Mysore P. Detection of undamped sub-synchronous oscillations of wind generators with series compensated lines. In: 47th annual minnesota power systems conference. Minnesota, USA; 2011.

[3] Irwin GD, Jindal AK, Isaacs AL. Sub-synchronous control interactions between Type 3 wind turbines and series compensated AC transmission systems. In: IEEE PES general meeting detroit. USA; 2011, p. 1–6.

[4] Xie X, Zhang X, Liu H, Liu H, Li Y, Zhang C. Characteristic analysis of sub-synchronous resonance in practical wind farms connected to series-compensated transmissions. IEEE Trans Energy Convers 2017;32:1117–26.

[5] Li Y, Fan L, Miao Z. Replicating real-world wind farm SSR events. IEEE Trans Power Deliv 2020;35(1):339–48.

[6] IEEE PES wind SSO task force wind energy systems sub-synchronous oscillations: events and modeling. Tech. rep. PES-TR80, USA: IEEE; 2020.

[7] Shair J, Xie X, Liu W, Li X, Li H. Modeling and stability analysis methods for investigating subsynchronous control interaction in large-scale wind power systems. Renew Sustain Energy Rev 2021;135(110420).

[8] Fan L, Zhu C, Miao Z, Hu M. Modal analysis of a DFIG-based wind farm interfaced with a series compensated network. IEEE Trans Energy Convers 2011;26(4):1010–20.

[9] Wang L, Xie X, Jiang Q, Liu H, Li Y, Liu H. Investigation of SSR in practical DFIG-based wind farms connected to a series-compensated power system. IEEE Trans Power Syst 2015;30(5):2772–9.

[10] Mohammadpour H, Santi E. SSR damping controller design and optimal placement in rotor-side and grid-side converters of series-compensated DFIG-based wind farm. IEEE Trans Sustain Energy 2015;6(2):388–99.
[11] León AE, Solsona JA. Sub-synchronous interaction damping control for DFIG wind turbines. IEEE Trans Power Syst 2015;30(1):419–28.
[12] Revel G, Alonso DM. Subsynchronous interactions in power networks with multiple DFIG-based wind farms. Electr Power Syst Res 2018;165:179–90.
[13] Du W, Wang Y, Wang H, Fu Q. Open-loop modal analysis to identify the SSO instability risk caused by grid connected wind farms. Int J Electr Power Energy Syst 2019;107:352–62.
[14] Liu J, Yao W, Wen J, Fang J, Jiang L, He H, Cheng S. Impact of power grid strength and PLL parameters on stability of grid-connected DFIG wind farm. IEEE Trans Sustain Energy 2020;11(1):545–57.
[15] Miao Z. Impedance-model-based SSR analysis for type 3 wind generator and series-compensated network. IEEE Trans Energy Convers 2012;27(4):984–91.
[16] Badrzadeh B, Sahni M, Zhou Y, Muthumuni D, Gole A. General methodology for analysis of sub-synchronous interaction in wind power plants. IEEE Trans Power Syst 2013;28(2):1858–69.
[17] Cheng Y, Sahni M, Muthumuni D, Badrzadeh B. Reactance scan crossover-based approach for investigating SSCI concerns for DFIG-based wind turbines. IEEE Trans. Power Del. 2013;28(2):742–51.
[18] Ren W, Larsen E. A refined frequency scan approach to subsynchronous control interaction (SSCI) study of wind farms. IEEE Trans Power Syst 2016;31(5):3904–12.
[19] Liu H, Xie X, Zhang C, Li Y, Liu H, Hu Y. Quantitative SSR analysis of series-compensated DFIG-based wind farms using aggregated RLC circuit model. IEEE Trans Power Syst 2017;32(1):474–83.
[20] Xu Y, Nian H, Wang T, Chen L, Zheng T. Frequency coupling characteristic modeling and stability analysis of doubly fed induction generator. IEEE Trans Energy Convers 2018;33(3):1475–86.
[21] Zhang C, Cai X, Molinas M, Rygg A. Frequency-domain modelling and stability analysis of a DFIG-based wind energy conversion system under non-compensated AC grids: impedance modelling effects and consequences on stability. IET Power Electron 2019;12(4):907–14.
[22] Zhan Y, Xie X, Liu H, Liu H, Li Y. Frequency-domain modal analysis of the oscillatory stability of power systems with high-penetration renewables. IEEE Trans Sustain Energy 2019;10(3):1534–43.
[23] Cheng Y, Huang SH, Rose J. A series capacitor based frequency scan method for SSR studies. IEEE Trans Power Deliv 2019;34(6):2135–44.
[24] Suriyaarachchi D, Annakkage U, Karawita C, Jacobson D. A procedure to study sub-synchronous interactions in wind integrated power systems. IEEE Trans Power Syst 2013;28(1):377–84.
[25] Zadehkhost PS, Howell F, Lin X, Wang L. Analyzing subsynchronous control interactions in large-scale power systems in frequency domain. In: 2017 IEEE PES general meeting. Chicago, USA; 2017, p. 1–5.
[26] Shair J, Xie X, Yan G. Mitigating subsynchronous control interaction in wind power systems: Existing techniques and open challenges. Renew Sustain Energy Rev 2019;108:330–46.
[27] Fan L, Kavasseri R, Miao Z, Zhu C. Modeling of DFIG-based wind farms for SSR analysis. IEEE Trans Power Deliv 2010;25(4):2073–82.
[28] Liu W, Xie X, Shair J, Li X. A nearly decoupled admittance model for grid-tied vscs under variable operating conditions. IEEE Trans Power Electron 2020;35(9):9380–9.
[29] Liu B, Li Z, Wang H, Dong X, Su W, Li G, Chen X, Tyrone F, Iu H, Lian X, Xiangdong L. Impedance modeling of DFIG wind farms with various rotor speeds and frequency coupling. IEEE Trans Circuit Syst II 2021;68(1):406–10.
[30] Dong W, Du W, Xie X, Wang HF. An approximate aggregated impedance model of a grid-connected wind farm for the study of small-signal stability. IEEE Trans Power Syst 2022;37(5).
[31] Zhan Y, Xie X, Chai W, Jiang Q. Security region analysis of sub/super-synchronous oscillations in wind power systems. In: 2021 IEEE 4th international conference on renewable energy and power engineering. REPE, 2021, p. 324–8.
[32] Liu W, Shair J, Wu S, Xie X. Oscillatory stability region analysis of black-box CIGs. IEEE Trans Power Electron 2022;37(8):8780–4.
[33] Liu W, Xie X, Shair J, Zhang J. Stability region analysis of grid-tied voltage sourced converters using variable operating point impedance model. IEEE Trans Power Syst 2022. In press.
[34] Cheah-Mane M, Sainz L, Liang J, Jenkins N, Ugalde-Loo CE. Criterion for the electrical resonance stability of offshore wind power plants connected through HVDC links. IEEE Trans Power Syst 2017;32(6):4579–89.
[35] Torquato R, Argüello A, Freitas W. Practical chart for harmonic resonance assessment of DFIG-based wind parks. IEEE Trans Power Del 2020;35(5):2233–42.
[36] Ke S, Lin T, Sheng Y, Du H, Li S, Chen B, Chen R, Li Q. Sub/super-synchronous oscillation oriented dominant controller parameters stability region based on Gerschgorin Disk Theorem. Appl Sci 2021;11:1205.
[37] Yu Y, Liu Y, Qin C, Yang T. Theory and method of power system integrated security region irrelevant to operation states: An introduction. Engineering 2020;6(7):754–77.
[38] Revel G, León AE, Alonso DM, Moiola JL. Multi-parameter bifurcation analysis of subsynchronous interactions in DFIG-based wind farms. Electr Power Syst Res 2016;140:643–52.
[39] Dhooge A, Govaerts W, Kuznetsov YA, Meijer H, Sautois B. New features of the software MatCont for bifurcation analysis of dynamical systems. Math Comput Model Dyn Syst 2008;14(2):147–71.
[40] Perez-Arriaga I, Verghese G, Schweppe F. Selective modal analysis with applications to electric power systems, Part I: Heuristic introduction. IEEE Trans Power Appl Syst 1982;PAS-101(9):3117–25.
[41] Verghese G, Perez-Arriaga I, Schweppe F. Selective modal analysis with applications to electric power systems, Part II: The dynamic stability problem. IEEE Trans Power Appl Syst 1982;PAS-101(9):3126–34.
[42] N. E. Costa G. Revel D. M. Alonso RD. Fernández Subsynchronous control interaction studies in DFIG-based wind farms using selective modal analysis. Int J Electr Power Energy Syst 2020;123:106291.
[43] Kuznetsov YA. Elements of applied bifurcation theory. Applied mathematical sciences, Springer; 2004.
[44] Kundur P. Power system stability and control. EPRI power system engineering series, McGraw-Hill; 1994.
[45] Govaerts WJF. Numerical methods for bifurcations of dynamical equilibria. Society for Industrial and Applied Mathematics; 2000.
[46] Broyden CG. A class of methods for solving nonlinear simultaneous equations. Math Comp 1965;19(92):577.
[47] Broyden CG. On the discovery of the good broyden method mathemtaical programming. Series B 2000;87:209–13.
[48] Hussein A, Chen K. On efficient methods for detecting Hopf bifurcation with applications to power system instability prediction. Int J Bifurcation Chaos 2003;13(5):1247–62.